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Name: _____

November 6, 2016 Sunday

Department/School: _____

SID: _____

University Physics II

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20pts.) Three point charges $2q$, q and $2q$ are restricted to the closed interval $-a \leq x \leq a$ on the x-axis.

(a) Determine the positions where the charges should be placed so that the potential energy of the system is a minimum.

(b) Supposing that the three point charges are located at the positions as determined in (a), find the electrical potential $V(x)$ at the point x on the x-axis where $|x| \gg a$, and approximate the result by expanding it up to the second order of $a/|x|$.

(c) For the same arrangement of the point charges in (b), find the approximate electrical field $E(x)$ at the point x on the x-axis (where $|x| \gg a$) up to the second order of $a/|x|$.

2.(20pts.) A pair of magnetic monopoles are placed at $\mathbf{r} = \mathbf{a}$ and $\mathbf{r} = -\mathbf{a}$, with the corresponding magnetic charges being g and $-g$, respectively.

(a) According to Dirac's hypothesis, write down the magnetic induction $\mathbf{B}$ at the point $\mathbf{r}$ ($\mathbf{r} \neq \pm\mathbf{a}$).

(b) The configuration can also be taken as a magnetic dipole when viewed from the far zone ($r \gg a, a = |\mathbf{a}|$), and the dipole moment can be defined as $\mathbf{p} = 2g\mathbf{a}$. Find the magnetic induction $\mathbf{B}$ produced by this dipole by expanding the expression of (a) up to the first order of a/r .

(c) Suppose there is another magnetic dipole moment $\boldsymbol{\mu}$ that is constituted of a small electric current loop in the magnetic field $\mathbf{B}$ of (b). The dipole moment $\boldsymbol{\mu}$ is located at the position $\mathbf{r}$ within the far zone. Find the torque $\mathbf{M}$ acted on the magnetic dipole $\boldsymbol{\mu}$ if $\mathbf{r} = x\mathbf{i}, \boldsymbol{\mu} = \mu\mathbf{j}$ and $\mathbf{a} = a\mathbf{k}$ where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the three base vectors of Cartesian system.

(Hint: $f(\mathbf{r} + \mathbf{a}) = f(\mathbf{r}) + \mathbf{a} \cdot \nabla f(\mathbf{r}) + \dots$, and in particular $\frac{1}{|\mathbf{r} + \mathbf{a}|^3} = \frac{1}{r^3} + \mathbf{a} \cdot \nabla \frac{1}{r^3} + \dots$ and $\nabla \frac{1}{r^3} = -\frac{3\mathbf{r}}{r^5}$.)

3. (20pts.) Three thin cylindrical shells are coaxially fixed together; their radii are R , $2R$, and $4R$, respectively. The inner and outer shells are both metallic and are both connected to the ground; the middle one is plastic, with areal charge density σ .
- (a) Find the areal charge density σ_1 of the inner shell.
- (b) Find the electric field $\mathbf{E}_1$ between the inner and the middle shells, and $\mathbf{E}_2$ between the middle and the outer ones.

4. (20pts.) Two parallel plates enclose a dielectric material that has a varying relative dielectric function: $\epsilon_r = \epsilon_r(x) = e^{\alpha x}$ where $\alpha > 0$ is a constant and x is the distance apart from the lower plate. Suppose that the areal charge densities of the upper and lower plates are σ and $-\sigma$ respectively.
- (a) What is the electric displacement $\mathbf{D}$ within the medium?
 - (b) What is the electric field $\mathbf{E}$ within the medium?
 - (c) If the two plates are of area S , and a distance d apart, find the energy U stored in this configuration.
 - (d) Find the volume charge density ρ distributing within the medium.

5. (20 pts.) A long solenoid of radius a , carrying n turns per unit length, is looped perpendicularly from outside by a closed conducting wire of electrical resistance R .

- (a) If the current I in the solenoid is increasing at a constant rate ($dI/dt = k$), what electrical current J flows in the loop?
- (b) If the current I in the solenoid is keeping constant but the solenoid is pulled out of the wire loop, turned around, and reinserted into it perpendicularly, what total charge Q that passes through the conducting loop during the process.

6. (20pts.) In free space, there exists a traveling electromagnetic wave $\mathbf{B}(x, y, z) = B_0 \cos(kz - \omega t) \mathbf{e}_y$, where $\mathbf{B}$ is the magnetic induction. Here, B_0 is a constant, x , y and z are Cartesian coordinates, and $\mathbf{e}_x$, $\mathbf{e}_y$ and $\mathbf{e}_z$ the corresponding unit vectors, t denotes the time, k the wave vector, and ω the frequency: $\omega = ck$ where c is the speed of light.

(a) Show that $\nabla \cdot \mathbf{B} = 0$.

(b) Supposing that the electric field $\mathbf{E}$ takes on the form: $\mathbf{E} = \mathbf{E}_0 \cos(kz - \omega t)$ where $\mathbf{E}_0$ is a constant vector, find $\mathbf{E}_0$ by the Maxwell equation $\nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}$.

(c) Find the Poynting vector $\mathbf{S}$ and its average $\langle \mathbf{S} \rangle$ over one period of time.